CivicFix AI — Final Evidence Addendum

Anonymized submission | Group 8 | 29 October 2026

1. Scope and research integrity

This document supplements the NYC 311 analysis. It reports an Istanbul point-rainfall case study, a Copernicus DEM / OpenStreetMap terrain-screening prototype, and a 25-item Turkish-language text pilot. It is not a validated flood forecast or a deployed municipal decision system. The team-member names and direct personal identifiers have been excluded from this delivery.

2. Evidence completed

Component	Result and interpretation
NYC 311 recurrence model	132,506 cleaned NYC 2025 complaints; ROC-AUC 0.672; PR-AUC 0.585; precision in top decile 71.6%. NYC recurrence, NOT Istanbul flood accuracy.
Istanbul precipitation	Supplied Open-Meteo single-coordinate series: 167.8 mm daily rainfall on 14 October 2026. This is not a citywide rainfall measurement.
Terrain and waterways	Copernicus DEM local slope / elevation heuristic combined with 99 mapped OSM waterway geometries in a limited footprint. Distance to mapped waterway is a proxy, not validated flood probability.
Turkish sample	25 anonymized paraphrases from user-provided posts: 7 attributed to Şikayetvar and 18 to Ekşi Sözlük. No synthetic additions in this final set. Original individual URLs/timestamps not verified.
GPT-6 classification pilot	25 text summaries classified in-chat with GPT-6; urgency accuracy 0.960; macro-F1 0.971 versus provisional project labels. Not a blind independent benchmark or external API test.

3. Turkish text pilot methodology

Inputs were de-identified summaries of user-supplied Turkish texts. Each was assigned an issue type and one urgency class: critical, high, routine, or not_actionable. GPT-6 predictions were produced in this ChatGPT session from the paraphrases. They were compared against provisional labels created during data preparation. Because the reference labels were not independently adjudicated, and the model was not tested blind on verified original posts, agreement is descriptive only and likely optimistic. No external LLM API was invoked.

Observed agreement: 24/25; accuracy 0.960; macro-F1 0.971. Class-wise precision, recall and F1, plus the confusion matrix, are in gpt6_pilot_metrics.json. Full row-level predictions are in gpt6_pilot_predictions_25.csv.

4. Privacy and source status

The deliverable stores only short anonymized paraphrases, coarse district/neighborhood labels and source-platform attributions. Names, personal phone numbers, number plates, street-level addresses, apartment/building numbers and user handles are not included in the test set. Place names are retained as geographic context; a small dataset cannot guarantee zero re-identification risk. The original source posts have not been independently authenticated. Statements about casualties, rescues, Tokyo rainfall or individual infrastructure incidents are treated as user claims, not verified facts.

5. Instructor feedback alignment

- Istanbul-specific rainfall: demonstrated at one Open-Meteo coordinate.
- Turkish classification: 25-text GPT-6 pilot now included, but without independently verified gold labels.
- Geospatial analysis: Copernicus DEM and mapped OSM waterways combined in the Istanbul pilot area.
- NYC–Istanbul transfer: common feature plan discussed; NYC DEM/OSM feature parity and external transfer validation not completed.
- Sentinel-1 event validation: not performed.
- Istanbul ground-truth flood validation: not performed; available posts lack individual verified provenance.

6. Limitations and future work

The 25 texts are a convenience sample, including non-actionable commentary. This pilot does not establish real-world precision of a municipal triage system. A subsequent study should collect dated and geolocated complaints with permission and verified source links, obtain independent multi-annotator labels, evaluate the LLM on original anonymized text, compute equivalent NYC/Istanbul geospatial features, and validate against observed inundation or authoritative event records. Sentinel-1 should be used only if image timing and urban geometry support reliable mapping.

7. Included files

Original NYC report (personal names redacted); source provenance; 25-item anonymized CSV; GPT-6 pilot predictions and metrics; Istanbul terrain/OSM image and metadata; reproducible geospatial script; this report in DOCX and PDF; README and privacy audit.